

منطق عمل الشبكة (Network Logic Overview) :

❖ داخل الوزارة (HQ) :

- شبكات المستخدمين (VLAN 10, 20, 30, 40) **معزولة تماماً عن بعضها البعض** - ، ولا يسمح لها بالتواصل إلا مع :
 - ✓ **الخوادم المركزية** (VLAN 50 - البيانات Data والدليل النشط AD)
 - ✓ **شبكة الإدارة المحلية** (VLAN 99 - لإدارة أجهزة الشبكة عن بُعد)
 - ✓ **شبكات السفارات** عبر النفق المشفر

❖ الأنفاق المشفرة (VPN Tunnels) :

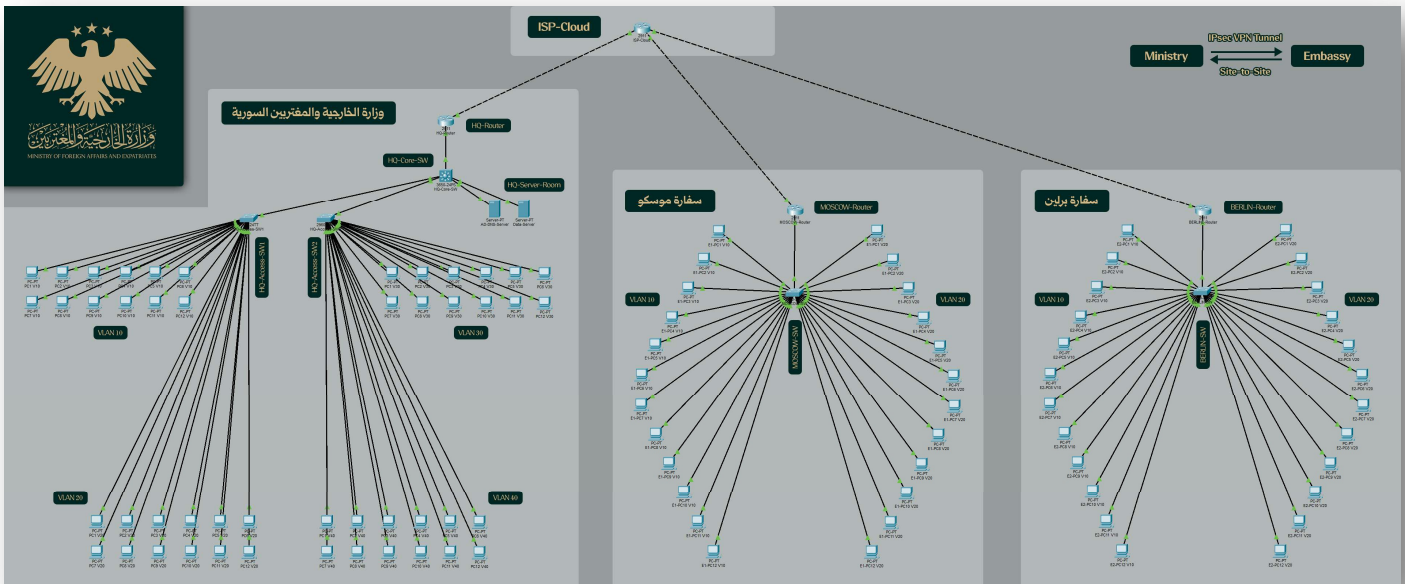
- بين الوزارة وكل سفارة (موسكو، برلين) نفق مؤمن بمعايير IPsec
- عبر النفق : **أي شبكة مستخدمين** في الوزارة (VLAN 10, 20, 30, 40) تستطيع التواصل مع **جميع الشبكات في السفارة** (VLAN 10, 20, 99) وكذلك الخوادم (VLAN 50) وشبكة الإدارة (VLAN 99)
- الاتصال بين المواقع **مؤمن بالكامل** ولا يخرج إلى الإنترنت العام دون تشفير إطلاقاً

❖ داخل السفارة الواحدة (Embassy) :

- شبكة البرمجة (VLAN 10) وشبكة المالية (VLAN 20) **معزولتان عن بعضهما محلياً** - ، لكن كلتيهما تصل إلى :
 - ✓ **شبكة الإدارة المحلية** (VLAN 99)
 - ✓ **شبكات الوزارة** عبر النفق المشفر

❖ الإنترنت العام :

- جميع المواقع تخرج إلى الإنترنت عبر **مسارها الافتراضي** إلى مزود الخدمة (ISP) دون المرور بالنفق، **باستثناء الحزم** التي تستوفي شروط التشفير المحددة مسبقاً (شروط المرور بالنفق)، فتلك تُمرّر تلقائياً عبر الأنفاق السيادية بين الوزارة وبعثاتها الخارجية والعكس.



ISP-Cloud

===== Basic Configuration & Security =====

```
enable secret ministry
no ip domain-lookup
hostname ISP-Cloud
!
line console 0
  password ministry
  login
!
line vty 0 4
  password ministry
  login
!
```

===== WAN Links to Remote Sites =====

```
interface Gi0/0
  description Link to HQ-Damascus
  ip address 10.1.0.2 255.255.255.252
  no shutdown
!
interface Gi0/1
  description Link to Embassy-Moscow
  ip address 10.2.0.2 255.255.255.252
  no shutdown
!
interface Gi0/2
  description Link to Embassy-Berlin
  ip address 10.3.0.2 255.255.255.252
  no shutdown
!
```

===== Static Routes =====

```
! HQ Routes
ip route 192.168.10.0 255.255.255.0 10.1.0.1
ip route 192.168.20.0 255.255.255.0 10.1.0.1
ip route 192.168.30.0 255.255.255.0 10.1.0.1
ip route 192.168.40.0 255.255.255.0 10.1.0.1
ip route 172.16.50.0 255.255.255.0 10.1.0.1
ip route 192.168.99.0 255.255.255.0 10.1.0.1
! Moscow Routes
ip route 192.169.10.0 255.255.255.0 10.2.0.1
ip route 192.169.20.0 255.255.255.0 10.2.0.1
ip route 192.169.99.0 255.255.255.0 10.2.0.1
! Berlin Routes
ip route 192.170.10.0 255.255.255.0 10.3.0.1
ip route 192.170.20.0 255.255.255.0 10.3.0.1
ip route 192.170.99.0 255.255.255.0 10.3.0.1
!
end
```

HQ-Router

===== Basic Configuration & Security =====

```
no ip domain-lookup
hostname HQ-Router
!
ip domain-name mofa.gov.sy
crypto key generate rsa 1024
username admin privilege 15 secret ministry
!
line console 0
  login local
!
line vty 0 4
  login local
  transport input ssh
  exec-timeout 5 0
!
```

===== WAN (to ISP) =====

```
interface Gi0/0
  description WAN Link to ISP-Cloud
  ip address 10.1.0.1 255.255.255.252
  no shutdown
!
```

===== Router-on-a-Stick =====

```
interface Gi0/1
  description Uplink to Core-Switch
  no shutdown
!
interface Gi0/1.10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.0
!
interface Gi0/1.20
  encapsulation dot1Q 20
  ip address 192.168.20.1 255.255.255.0
!
interface Gi0/1.30
  encapsulation dot1Q 30
  ip address 192.168.30.1 255.255.255.0
!
interface Gi0/1.40
  encapsulation dot1Q 40
  ip address 192.168.40.1 255.255.255.0
!
interface Gi0/1.50
  encapsulation dot1Q 50
  ip address 172.16.50.1 255.255.255.0
!
interface Gi0/1.99
  encapsulation dot1Q 99
  ip address 192.168.99.1 255.255.255.0
!
```

===== DHCP =====

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp excluded-address 192.168.30.1 192.168.30.10
ip dhcp excluded-address 192.168.40.1 192.168.40.10
ip dhcp excluded-address 172.16.50.1 172.16.50.10
!
ip dhcp pool VLAN10-Pool
  network 192.168.10.0 255.255.255.0
  default-router 192.168.10.1
  dns-server 172.16.50.10
!
ip dhcp pool VLAN20-Pool
  network 192.168.20.0 255.255.255.0
  default-router 192.168.20.1
  dns-server 172.16.50.10
!
ip dhcp pool VLAN30-Pool
  network 192.168.30.0 255.255.255.0
  default-router 192.168.30.1
  dns-server 172.16.50.10
```

```

!
ip dhcp pool VLAN40-Pool
 network 192.168.40.0 255.255.255.0
 default-router 192.168.40.1
 dns-server 172.16.50.10
!
ip dhcp pool VLAN50-Pool
 network 172.16.50.0 255.255.255.0
 default-router 172.16.50.1
 dns-server 172.16.50.10
!

```

===== Static Route =====

```

ip route 0.0.0.0 0.0.0.0 10.1.0.2
!

```

===== ACL =====

```

access-list 101 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 101 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 101 deny ip 192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255
access-list 101 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 101 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 101 deny ip 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255
access-list 101 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 101 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 101 deny ip 192.168.30.0 0.0.0.255 192.168.40.0 0.0.0.255
access-list 101 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 101 deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 101 deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 101 permit ip any 172.16.50.0 0.0.0.255
access-list 101 permit ip 172.16.50.0 0.0.0.255 any
access-list 101 permit ip any 192.168.99.0 0.0.0.255
access-list 101 permit ip 192.168.99.0 0.0.0.255 any
access-list 101 permit ip any any
!
interface Gi0/1.10
 ip access-group 101 in
!
interface Gi0/1.20
 ip access-group 101 in
!
interface Gi0/1.30
 ip access-group 101 in
!
interface Gi0/1.40
 ip access-group 101 in
!
interface Gi0/1.50
 ip access-group 101 in
!
interface Gi0/1.99
 ip access-group 101 in
!

```

===== Global IPsec VPN Policy =====

```

License boot module c2900 technology-package securityk9

crypto isakmp policy 10
 encr aes 256
 hash sha
 authentication pre-share
 group 5
 lifetime 86400
!
crypto ipsec transform-set AES-SHA esp-aes 256 esp-sha-hmac
!

```

===== VPN Tunnel : Moscow =====

```
crypto isakmp key MOFA_Moscow_Key address 10.2.0.1
!
access-list 110 permit ip 192.168.10.0 0.0.0.255 192.169.10.0 0.0.0.255
access-list 110 permit ip 192.168.10.0 0.0.0.255 192.169.20.0 0.0.0.255
access-list 110 permit ip 192.168.10.0 0.0.0.255 192.169.99.0 0.0.0.255
access-list 110 permit ip 192.168.20.0 0.0.0.255 192.169.10.0 0.0.0.255
access-list 110 permit ip 192.168.20.0 0.0.0.255 192.169.20.0 0.0.0.255
access-list 110 permit ip 192.168.20.0 0.0.0.255 192.169.99.0 0.0.0.255
access-list 110 permit ip 192.168.30.0 0.0.0.255 192.169.10.0 0.0.0.255
access-list 110 permit ip 192.168.30.0 0.0.0.255 192.169.20.0 0.0.0.255
access-list 110 permit ip 192.168.30.0 0.0.0.255 192.169.99.0 0.0.0.255
access-list 110 permit ip 192.168.40.0 0.0.0.255 192.169.10.0 0.0.0.255
access-list 110 permit ip 192.168.40.0 0.0.0.255 192.169.20.0 0.0.0.255
access-list 110 permit ip 192.168.40.0 0.0.0.255 192.169.99.0 0.0.0.255
access-list 110 permit ip 172.16.50.0 0.0.0.255 192.169.10.0 0.0.0.255
access-list 110 permit ip 172.16.50.0 0.0.0.255 192.169.20.0 0.0.0.255
access-list 110 permit ip 172.16.50.0 0.0.0.255 192.169.99.0 0.0.0.255
access-list 110 permit ip 192.168.99.0 0.0.0.255 192.169.10.0 0.0.0.255
access-list 110 permit ip 192.168.99.0 0.0.0.255 192.169.20.0 0.0.0.255
access-list 110 permit ip 192.168.99.0 0.0.0.255 192.169.99.0 0.0.0.255
!
crypto map HQ-VPN 10 ipsec-isakmp
 set peer 10.2.0.1
 set transform-set AES-SHA
 match address 110
!
```

===== VPN Tunnel : Berlin =====

```
crypto isakmp key MOFA_Berlin_Key address 10.3.0.1
!
access-list 120 permit ip 192.168.10.0 0.0.0.255 192.170.10.0 0.0.0.255
access-list 120 permit ip 192.168.10.0 0.0.0.255 192.170.20.0 0.0.0.255
access-list 120 permit ip 192.168.10.0 0.0.0.255 192.170.99.0 0.0.0.255
access-list 120 permit ip 192.168.20.0 0.0.0.255 192.170.10.0 0.0.0.255
access-list 120 permit ip 192.168.20.0 0.0.0.255 192.170.20.0 0.0.0.255
access-list 120 permit ip 192.168.20.0 0.0.0.255 192.170.99.0 0.0.0.255
access-list 120 permit ip 192.168.30.0 0.0.0.255 192.170.10.0 0.0.0.255
access-list 120 permit ip 192.168.30.0 0.0.0.255 192.170.20.0 0.0.0.255
access-list 120 permit ip 192.168.30.0 0.0.0.255 192.170.99.0 0.0.0.255
access-list 120 permit ip 192.168.40.0 0.0.0.255 192.170.10.0 0.0.0.255
access-list 120 permit ip 192.168.40.0 0.0.0.255 192.170.20.0 0.0.0.255
access-list 120 permit ip 192.168.40.0 0.0.0.255 192.170.99.0 0.0.0.255
access-list 120 permit ip 172.16.50.0 0.0.0.255 192.170.10.0 0.0.0.255
access-list 120 permit ip 172.16.50.0 0.0.0.255 192.170.20.0 0.0.0.255
access-list 120 permit ip 172.16.50.0 0.0.0.255 192.170.99.0 0.0.0.255
access-list 120 permit ip 192.168.99.0 0.0.0.255 192.170.10.0 0.0.0.255
access-list 120 permit ip 192.168.99.0 0.0.0.255 192.170.20.0 0.0.0.255
access-list 120 permit ip 192.168.99.0 0.0.0.255 192.170.99.0 0.0.0.255
!
crypto map HQ-VPN 20 ipsec-isakmp
 set peer 10.3.0.1
 set transform-set AES-SHA
 match address 120
!
```

===== Crypto Map on WAN =====

```
interface Gi0/0
 crypto map HQ-VPN
!
end
```

HQ-Core-SW

===== Basic Configuration & Security =====

```
hostname HQ-Core-SW
!
ip domain-name mofa.gov.sy
crypto key generate rsa 1024
username admin privilege 15 secret ministry
line console 0
  login local
!
line vty 0 4
  login local
  transport input ssh
  exec-timeout 5 0
!
```

===== VLANs Definition =====

```
vlan 10
  name Programming-DigitalTransformation
vlan 20
  name Finance-Budget
vlan 30
  name HR-AdministrativeAffairs
vlan 40
  name Legal-InternationalTreaties
vlan 50
  name Server-Room
vlan 99
  name Management
!
```

===== Spanning-Tree =====

```
spanning-tree mode rapid-pvst
spanning-tree vlan 10,20,30,40,50,99 root primary
spanning-tree portfast default
!
```

===== (Access - Server-Room) =====

```
interface Gil/0/1
  description Data-Server
  switchport mode access
  switchport access vlan 50
!
interface Gil/0/2
  description AD-DNS-Server
  switchport mode access
  switchport access vlan 50
!
```

===== (Trunk - Sub-Switches) =====

```
interface Gil/0/11
  description Trunk to HQ-Access-SW1 (VLAN 10,20)
  switchport mode trunk
!
interface Gil/0/12
  description Trunk to HQ-Access-SW2 (VLAN 30,40)
  switchport mode trunk
!
```

===== (Trunk - WAN) =====

```
interface Gil/1/1
  description Uplink to HQ-Router
  switchport mode trunk
!
```

===== Management =====

```
interface Vlan 99
  ip address 192.168.99.2 255.255.255.0
  no shutdown
ip default-gateway 192.168.99.1
!
end
```

HQ-Access-SW1

===== Basic Configuration & Security =====

```
hostname HQ-Access-SW1
!
ip domain-name mofa.gov.sy
crypto key generate rsa 1024
username admin privilege 15 secret ministry
line console 0
  login local
line vty 0 4
  login local
  transport input ssh
  exec-timeout 5 0
!
```

===== VLANs Definition =====

```
vlan 10
  name Programming-DigitalTransformation
vlan 20
  name Finance-Budget
vlan 99
  name Management
!
```

===== Spanning-Tree =====

```
spanning-tree mode rapid-pvst
spanning-tree portfast default
!
```

===== (Access - Computers) =====

```
interface range Fa0/1-12
  switchport mode access
  switchport access vlan 10
!
interface range Fa0/13-24
  switchport mode access
  switchport access vlan 20
!
```

===== (Trunk - Core-Switch) =====

```
interface Gi0/1
  description Uplink to HQ-Core-SW
  switchport mode trunk
!
```

===== Management =====

```
interface Vlan 99
  ip address 192.168.99.10 255.255.255.0
  no shutdown
ip default-gateway 192.168.99.1
!
end
```

HQ-Access-SW2

===== Basic Configuration & Security =====

```
hostname HQ-Access-SW2
!
ip domain-name mofa.gov.sy
crypto key generate rsa 1024
username admin privilege 15 secret ministry
line console 0
  login local
line vty 0 4
  login local
  transport input ssh
  exec-timeout 5 0
!
```

===== VLANs Definition =====

```
vlan 30
  name HR-AdministrativeAffairs
vlan 40
  name Legal-InternationalTreaties
vlan 99
  name Management
!
```

===== Spanning-Tree =====

```
spanning-tree mode rapid-pvst
spanning-tree portfast default
!
```

===== (Access - Computers) =====

```
interface range Fa0/1-12
  switchport mode access
  switchport access vlan 30
!
interface range Fa0/13-24
  switchport mode access
  switchport access vlan 40
!
```

===== (Trunk - Core-Switch) =====

```
interface Gi0/1
  description Uplink to HQ-Core-SW
  switchport mode trunk
!
```

===== Management =====

```
interface Vlan 99
  ip address 192.168.99.11 255.255.255.0
  no shutdown
ip default-gateway 192.168.99.1
!
end
```


MOSCOW-Router

===== Basic Configuration & Security =====

```
hostname MOSCOW-Router
ip domain-name moscow.mofa.gov.sy
crypto key generate rsa 1024
username admin privilege 15 secret ministry
!
line console 0
  login local
line vty 0 4
  login local
  transport input ssh
  exec-timeout 5 0
!
```

===== WAN (to ISP) =====

```
interface Gi0/0
  ip address 10.2.0.1 255.255.255.252
  no shutdown
!
```

===== Router-on-a-Stick =====

```
interface Gi0/1
  no shutdown
!
interface Gi0/1.10
  encapsulation dot1Q 10
  ip address 192.169.10.1 255.255.255.0
!
interface Gi0/1.20
  encapsulation dot1Q 20
  ip address 192.169.20.1 255.255.255.0
!
interface Gi0/1.99
  encapsulation dot1Q 99
  ip address 192.169.99.1 255.255.255.0
!
```

===== DHCP =====

```
ip dhcp excluded-address 192.169.10.1 192.169.10.10
ip dhcp excluded-address 192.169.20.1 192.169.20.10
!
ip dhcp pool VLAN10-Moscow
  network 192.169.10.0 255.255.255.0
  default-router 192.169.10.1
  dns-server 172.16.50.10
!
ip dhcp pool VLAN20-Moscow
  network 192.169.20.0 255.255.255.0
  default-router 192.169.20.1
  dns-server 172.16.50.10
!
```

===== ACL =====

```
access-list 101 deny ip 192.169.10.0 0.0.0.255 192.169.20.0 0.0.0.255
access-list 101 deny ip 192.169.20.0 0.0.0.255 192.169.10.0 0.0.0.255
access-list 101 permit ip 192.169.99.0 0.0.0.255 any
access-list 101 permit ip any 192.169.99.0 0.0.0.255
access-list 101 permit ip any any
!
```

```

interface Gi0/1.10
  ip access-group 101 in
!
interface Gi0/1.20
  ip access-group 101 in
!
interface Gi0/1.99
  ip access-group 101 in
!

```

===== Static Route =====

```

ip route 0.0.0.0 0.0.0.0 10.2.0.2
!

```

===== Global IPsec VPN Policy =====

```

License boot module c2900 technology-package securityk9

```

```

crypto isakmp policy 10
  encr aes 256
  hash sha
  authentication pre-share
  group 5
  lifetime 86400
crypto ipsec transform-set AES-SHA esp-aes 256 esp-sha-hmac
!

```

===== VPN Tunnel : HQ =====

```

crypto isakmp key MOFA_Moscow_Key address 10.1.0.1
!
access-list 110 permit ip 192.169.10.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 110 permit ip 192.169.10.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 110 permit ip 192.169.10.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 110 permit ip 192.169.10.0 0.0.0.255 192.168.40.0 0.0.0.255
access-list 110 permit ip 192.169.10.0 0.0.0.255 172.16.50.0 0.0.0.255
access-list 110 permit ip 192.169.10.0 0.0.0.255 192.168.99.0 0.0.0.255
access-list 110 permit ip 192.169.20.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 110 permit ip 192.169.20.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 110 permit ip 192.169.20.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 110 permit ip 192.169.20.0 0.0.0.255 192.168.40.0 0.0.0.255
access-list 110 permit ip 192.169.20.0 0.0.0.255 172.16.50.0 0.0.0.255
access-list 110 permit ip 192.169.20.0 0.0.0.255 192.168.99.0 0.0.0.255
access-list 110 permit ip 192.169.99.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 110 permit ip 192.169.99.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 110 permit ip 192.169.99.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 110 permit ip 192.169.99.0 0.0.0.255 192.168.40.0 0.0.0.255
access-list 110 permit ip 192.169.99.0 0.0.0.255 172.16.50.0 0.0.0.255
access-list 110 permit ip 192.169.99.0 0.0.0.255 192.168.99.0 0.0.0.255
!
crypto map EMB-VPN 10 ipsec-isakmp
  set peer 10.1.0.1
  set transform-set AES-SHA
  match address 110
!

```

===== Crypto Map on WAN =====

```

interface Gi0/0
  crypto map EMB-VPN
!
end

```

MOSCOW-SW

===== Basic Configuration & Security =====

```
hostname MOSCOW-SW
!
ip domain-name moscow.mofa.gov.sy
crypto key generate rsa 1024
username admin privilege 15 secret ministry
line console 0
  login local
line vty 0 4
  login local
  transport input ssh
  exec-timeout 5 0
!
```

===== VLANs Definition =====

```
vlan 10
  name Programming
vlan 20
  name Finance
vlan 99
  name Management
!
```

===== Spanning-Tree =====

```
spanning-tree mode rapid-pvst
spanning-tree portfast default
!
```

===== (Access - Computers) =====

```
interface range Fa0/1-12
  switchport mode access
  switchport access vlan 10
!
interface range Fa0/13-24
  switchport mode access
  switchport access vlan 20
!
```

===== (Trunk - WAN) =====

```
interface Gi0/1
  description Uplink to MOSCOW-Router
  switchport mode trunk
!
```

===== Management =====

```
interface Vlan 99
  ip address 192.169.99.2 255.255.255.0
  no shutdown
ip default-gateway 192.169.99.1
!
end
```

BERLIN-Router

===== Basic Configuration & Security =====

```
hostname BERLIN-Router
ip domain-name berlin.mofa.gov.sy
crypto key generate rsa 1024
username admin privilege 15 secret ministry
!
line console 0
  login local
line vty 0 4
  login local
  transport input ssh
  exec-timeout 5 0
!
```

===== WAN (to ISP) =====

```
interface Gi0/0
  ip address 10.3.0.1 255.255.255.252
  no shutdown
!
```

===== Router-on-a-Stick =====

```
interface Gi0/1
  no shutdown
!
interface Gi0/1.10
  encapsulation dot1Q 10
  ip address 192.170.10.1 255.255.255.0
!
interface Gi0/1.20
  encapsulation dot1Q 20
  ip address 192.170.20.1 255.255.255.0
!
interface Gi0/1.99
  encapsulation dot1Q 99
  ip address 192.170.99.1 255.255.255.0
!
```

===== DHCP =====

```
ip dhcp excluded-address 192.170.10.1 192.170.10.10
ip dhcp excluded-address 192.170.20.1 192.170.20.10
!
ip dhcp pool VLAN10-Berlin
  network 192.170.10.0 255.255.255.0
  default-router 192.170.10.1
  dns-server 172.16.50.10
!
ip dhcp pool VLAN20-Berlin
  network 192.170.20.0 255.255.255.0
  default-router 192.170.20.1
  dns-server 172.16.50.10
!
```

===== ACL =====

```
access-list 101 deny ip 192.170.10.0 0.0.0.255 192.170.20.0 0.0.0.255
access-list 101 deny ip 192.170.20.0 0.0.0.255 192.170.10.0 0.0.0.255
access-list 101 permit ip 192.170.99.0 0.0.0.255 any
access-list 101 permit ip any 192.170.99.0 0.0.0.255
access-list 101 permit ip any any
```

```

interface GigabitEthernet0/1.10
  ip access-group 101 in
!
interface Gi0/1.20
  ip access-group 101 in
!
interface Gi0/1.99
  ip access-group 101 in
!

```

===== Static Route =====

```

ip route 0.0.0.0 0.0.0.0 10.3.0.2
!

```

===== Global IPsec VPN Policy =====

```

License boot module c2900 technology-package securityk9

```

```

crypto isakmp policy 10
  encr aes 256
  hash sha
  authentication pre-share
  group 5
  lifetime 86400
crypto ipsec transform-set AES-SHA esp-aes 256 esp-sha-hmac
!

```

===== VPN Tunnel : HQ =====

```

crypto isakmp key MOFA_Berlin_Key address 10.1.0.1
!
access-list 110 permit ip 192.170.10.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 110 permit ip 192.170.10.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 110 permit ip 192.170.10.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 110 permit ip 192.170.10.0 0.0.0.255 192.168.40.0 0.0.0.255
access-list 110 permit ip 192.170.10.0 0.0.0.255 172.16.50.0 0.0.0.255
access-list 110 permit ip 192.170.10.0 0.0.0.255 192.168.99.0 0.0.0.255
access-list 110 permit ip 192.170.20.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 110 permit ip 192.170.20.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 110 permit ip 192.170.20.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 110 permit ip 192.170.20.0 0.0.0.255 192.168.40.0 0.0.0.255
access-list 110 permit ip 192.170.20.0 0.0.0.255 172.16.50.0 0.0.0.255
access-list 110 permit ip 192.170.20.0 0.0.0.255 192.168.99.0 0.0.0.255
access-list 110 permit ip 192.170.99.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 110 permit ip 192.170.99.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 110 permit ip 192.170.99.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 110 permit ip 192.170.99.0 0.0.0.255 192.168.40.0 0.0.0.255
access-list 110 permit ip 192.170.99.0 0.0.0.255 172.16.50.0 0.0.0.255
access-list 110 permit ip 192.170.99.0 0.0.0.255 192.168.99.0 0.0.0.255
!
crypto map EMB-VPN 10 ipsec-isakmp
  set peer 10.1.0.1
  set transform-set AES-SHA
  match address 110
!

```

===== Crypto Map on WAN =====

```

interface Gi0/0
  crypto map EMB-VPN
!
end

```

BERLIN-SW

===== Basic Configuration & Security =====

```
hostname BERLIN-SW
!
ip domain-name berlin.mofa.gov.sy
crypto key generate rsa 1024
username admin privilege 15 secret ministry
line console 0
  login local
line vty 0 4
  login local
  transport input ssh
  exec-timeout 5 0
!
```

===== VLANs Definition =====

```
vlan 10
  name Programming
vlan 20
  name Finance
vlan 99
  name Management
!
```

===== Spanning-Tree =====

```
spanning-tree mode rapid-pvst
spanning-tree portfast default
!
```

===== (Access - Computers) =====

```
interface range Fa0/1-12
  switchport mode access
  switchport access vlan 10
!
interface range Fa0/13-24
  switchport mode access
  switchport access vlan 20
!
```

===== (Trunk - WAN) =====

```
interface Gi0/1
  switchport mode trunk
!
```

===== Management =====

```
interface Vlan 99
  ip address 192.170.99.2 255.255.255.0
  no shutdown
ip default-gateway 192.170.99.1
!
end
```